Rakan Eyad

Mechatronic Engineer | Focus on Aerospace, GNC, and Embedded Systems

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Education

University of Sydney — Bachelor of Engineering (Honours)
Mechatronic Engineering, 2022 – Present
Current EWAM in Final Semester: 76% (Distinction)
Relevant Coursework: Advanced Control Systems, Radar Systems & Antenna Patterns, Sensors & Signals, Computer Vision, Robotics, Signal Processing, Embedded Systems, CAD

Project Experience

Planetary Rover Autonomous Navigation Thesis (2026 - Present)

- Developing a reinforcement learning algorithm for the autonomous navigation of a planetary rover.
- Utilizing NVIDIA Isaac Sim to build high-fidelity physics simulations and model complex traversal environments.
- Integrating a rover CAD design into the simulation pipeline to test and validate state estimation and control logic.

Autonomous Target Tracking & Air Defence System (Personal Project)

- Designed and built a compact tracking system utilizing a laser to autonomously lock onto and track moving targets.
- Applied advanced control theory, sensor calibration, and signal filtering for rapid actuation and precise targeting.
- Currently expanding the system's capabilities to include predictive projectile targeting based on moving target velocity and trajectory.

Satellite Refueling CAD Design (ANT61) (2024)

- Designed a prototype latching mechanism for mechanical autonomous satellite refueling using Fusion360.

Assistive Robot Development (2024)

- Developed a Bluetooth-controlled robot using an FPGA board and Verilog to follow and respond to dynamic commands.
- Integrated diverse sensors and closed-loop control systems for autonomous hardware operation

Autonomous Robot with TurtleBot and ROS2 (2024)

- Programmed an autonomous TurtleBot using ROS2 to identify, count, and interact items using QR codes.
- Applied logic, sensor fusion, and path-finding algorithms applicable to autonomous robotic platforms.

Technical Skills

Core Competencies: Guidance, Navigation & Control (GNC), Sensor Integration, Signal Processing, Autonomous Pathfinding, Hardware-in-the-Loop (HIL) Simulation, Embedded Systems, Computer Vision
Software: SolidWorks, Fusion360, MATLAB, Excel, Word, PowerPoint
Programming: Python, C, C++, Arduino, Assembly, SystemVerilog, Verilog, R
Platforms: ROS2, FPGA, Embedded Systems, Isaac Sim, Isaac Lab

Transferable Skills

Leadership: Successfully led technical teams from planning to delivery.
Collaboration: Worked across teams in both industry (CloudAk, Escape Room) and uni projects.
Communication: Experienced in client interaction and technical presentations.
Adaptability: Learned ROS2, FPGA workflows, Isaac Sim, Isaac Lab and CAD tools independently.

Industry Experience

ALBON — System Engineer
November 2024 – January 2026
- Developed and integrated hardware, software, and mechanical components for a prototype control system, including transitioning the microcontroller architecture to a Raspberry Pi for enhanced real-time data processing.
- Integrated sensor arrays, and engineered the system's electrical junction box to optimize power consumption and ensure reliable hardware performance.

CloudAk — Testing Technician Intern
May 2021 – August 2021
- Supported server system testing and maintenance in data center environments.

Escape The Room — Escape Room Engineer Intern
September 2021 – November 2021
- Designed mechanical puzzles and implemented engineering concepts in real-world client products.

References

Available on request.